



Optical characterization of novel matrix glasses based on a CdO:ZnO:V₂O₅ ternary system

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ABSTRACT

Based on the adequate proportion in wt.% G₁(70:10:20), G₂(50:30:20), G₃(60:20:20) and G₄(70:20:10) of the oxides CdO:ZnO:V₂O₅, new types of glassy matrices were fabricated. By exploring within the Gibbs's mixture triangle, 36 combinations of the CdO:ZnO:V₂O₅ reactants were examined and some of these combinations within this triangle yielded transparent and brilliant glasses. The X-ray patterns indicated that the glassy matrices were totally amorphous. In the IR spectra for the G₁–G₄ glasses, recorded at room temperature, two different bands were observed. One band was localized at 520 cm^{−1} and was assigned to the presence of unmelted ZnO, one of the starting materials. Another wide band in the interval of 1200–630 cm^{−1} was assigned to the presence of CdV₂O₆, Cd₂V₂O₇, ZnV₂O₆ and Zn₂V₂O₇ compounds. The optical absorption allowed the evaluation of an E_g in the region of 4.2–5.4 eV for this group of glasses, and by Raman spectroscopy, changes in vibrations of the V=O bond appeared.

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1. Introduction

The optical properties of spectral transmittance, refractive index, and homogeneity are very important features in glasses and can be analyzed and exploited. Studies of vitrifications with vanadate systems have been carried out using several binary systems, and these studies have examined structural changes and coordination changes [1–4]. Additionally, these novel substances were used to prepare structures like nanotubes, macroporous nanostructures, and thin films [5–8], and all these structures were fabricated based on V₂O₅. V₂O₅ has proven to be suitable for consideration as a component in the fabrication of a new generation of glasses, and these new materials hold great promise for fabrication of structures that could be used in future devices.

In this preliminary research, novel glass matrices were fabricated from unexpected compositions in ternary systems (Fig. 1) with the oxides CdO, ZnO, and V₂O₅, since in this kind of glasses, the glass formator (V₂O₅) is in smaller proportion than the oxides modifiers. The obtained glasses possess great hardness and good reflectance. In this initial report, however, the hardness was not

determined by analytical methods because these new glass materials were very difficult to fragment.

2. Experimental

The glasses were fabricated by using a conventional melt quenching method in a platinum crucible in the temperature range of 800–1200 °C using an electrically heated Thermolyne furnace model 4800 in an air atmosphere. By dividing each of the reactants into ten different combinations of wt.% (Fig. 1), we prepared and examined all the resulting combinations within the Gibbs's triangle. The ternary mixtures were melted, but just the resulting glasses with similar characteristics were studied in this preliminary investigation. The four glasses (G₁–G₄) with different nominal compositions of the V₂O₅, CdO, and ZnO starting compounds were analyzed (Table 1). X-ray powder diffraction patterns were determined employing a Siemens D5000 diffractometer and using the Cu Kα line. The IR spectra were recorded at room temperature using a Bruker Optik GmbH model Vertex 70 spectrometer in the range of 1200–500 cm^{−1}. The absorption spectra were determined with a Unicam system with a spectral range from 290 to 1100 nm. The Raman spectra were measured with a micro-Raman Lab-Ram Dilor system with a 514 nm excitation line. In energy dispersive

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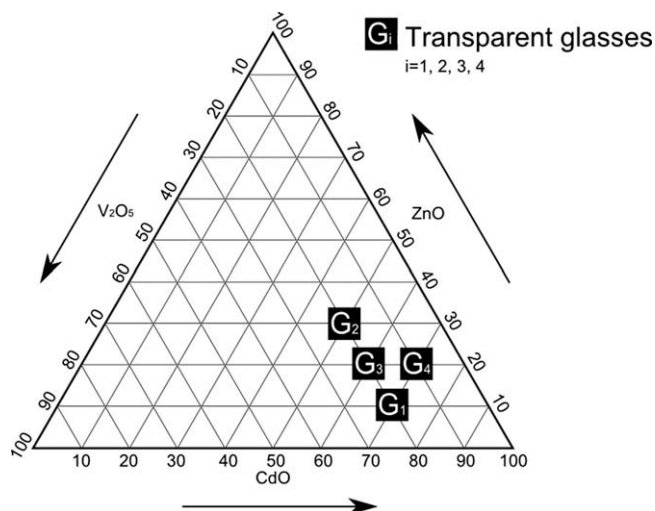


Fig. 1. Gibbs's ternary diagram system showing the CdO:ZnO:V₂O₅ compositions for the transparent glasses G₁–G₄.

Table 1
Nominal compositions for the obtained glasses.

Sample	CdO (wt.%)	ZnO (wt.%)	V ₂ O ₅ (wt.%)
G ₁	70	10	20
G ₂	50	30	20
G ₃	60	20	20
G ₄	70	20	10

spectroscopy (EDS) studies, the chemical composition analyses were determined utilizing an 1100/1110 EDX system from Noran Instruments.

3. Results and discussion

The amorphous nature of the G₁–G₄ glasses was confirmed by X-ray powder diffraction patterns, which indicated that there was no crystalline structure in the samples. The calculated data for the semi-quantitative EDS analyses of the G₁–G₄ glasses (Table 2) suggested that oxygen was present in the highest proportion, with the next highest proportion being Cd, and the least prevalent components being V and Zn. These results are in agreement with the nominal compositions of the glass mixtures. The components of each system are the oxides of the metals, and thus, oxygen is present as the major element, with Cd being intermediate, and with V and Zn making the lowest contributions.

The IR spectra for the G₁–G₄ glasses, depicted in Fig. 2, show a band in each spectrum at ~520 cm⁻¹. This band can be related to residual ZnO in these glassy matrices, which could be present due to ZnO that was not totally melted. The IR spectrum of the starting chemical ZnO used in the fabrication of the glasses (Fig. 2, inset) showed a remarkably strong absorption at ~520 cm⁻¹, which was also observed for the individual glasses,

Table 2
EDS measurements of the compositions for the obtained glasses.

Sample	Cd (at.%)	Zn (at.%)	V (at.%)	O (at.%)
G ₁	29.45	1.0	7.93	61.12
G ₂	21.65	2.17	10.3	65.88
G ₃	18.74	2.72	7.64	70.9
G ₄	22.6	1.1	9.25	67.04

although with less intensity. The melting point of ZnO is 1975 °C and the G₁–G₄ glasses were fabricated in the 800–1200 °C range, and thus, the ZnO may not have fully melted, which may account for the residual levels of this reactant in these glassy matrices.

Another wide band was present in these spectra at 1200–630 cm⁻¹ (Fig. 2). The spectra exhibited minima at 920, 940, 850, and 960 cm⁻¹ for glasses G₁, G₂, G₃, and G₄, respectively. In the binary systems with higher content of CdO and ZnO, bands at 970–910 and 850 cm⁻¹ were correlated with the presence of CdV₂O₆, ZnV₂O₆ (metavanadate) and Cd₂V₂O₇, Zn₂V₂O₇ (pyrovanadate) groups. Specifically, the first interval of the wavenumber was assigned to an elongation of the V=O bonds affected by the influence of Cd²⁺ or Zn²⁺ ions, and the frequency at 850 cm⁻¹ was associated with a symmetric vibration of VO₃ groups [1,4]. For other regions of the Gibbs's triangle, different from those defining the glasses G₁–G₄, the glass compositions did not produce transparent and reflecting samples. Because of the fixed glass formation region, it was not possible to perform a band follow-up as in binary systems [1,4]. However, we assume that the bands identified correspond to the above described compounds in this ternary system.

The optical absorption spectra for the glasses G₁–G₄, as a function of the $h\nu$ (Fig. 3), where h is Planck's constant and ν is the frequency of the light, were acquired with steps of 1 nm/s, taking into account an average-thickness of 3 mm, and assuming that the E_g for G₁–G₄ glasses was in the 4.2–5.4 eV range. The $(\alpha h\nu)^2$ vs. $h\nu$ relation for calculating the E_g was used as previously described, based on photoacoustic spectroscopy of glassy matrices of the same type [9]. In this case, the optical absorption was used directly because the transparency of the glasses allowed the measurement of the optical density. When vitrification was carried out by semiconductor, the band gaps for V₂O₅, CdO and ZnO in crystalline phase were 2.7, 2.5, and 3.3 eV respectively [10–12], but the band gaps values for each glass found after the vitrification were higher. As indicated by IR spectroscopy, these glasses are constituted mainly of CdV₂O₆, Cd₂V₂O₇; and ZnV₂O₆, Zn₂V₂O₇ compounds which could cause the elevated values of E_g . The difference between the E_g values found for the different glasses may be due to the different proportions of the CdV₂O₆, Cd₂V₂O₇, ZnV₂O₆, and Zn₂V₂O₇ compounds present in each glass.

The Raman spectra for the G₁–G₄ glasses were identical, as illustrated by a representative spectrum in Fig. 4. The spectra also indicated that the G₁–G₄ glasses were homogenous throughout, since the Raman spectra were recorded in different places in the samples. As reported by Maniu et al. [13] for a binary system where the content of V₂O₅ was 50%, a band at 935–910 cm⁻¹ was assigned to stretching vibrations of VO₃ units of the metavanadate groups and a band at 825–810 cm⁻¹ was assigned to vibrations of bridges of pyrovanadates groups. In the glasses fabricated with the present ternary system, the wt.% of V₂O₅ was varied between 10% and 20% and their maxima were localized at 840 cm⁻¹. However, the signal registered should be integrated using the vibrations that were previously defined, since the formation of such groups was also corroborated by IR spectroscopy. In these measurements, for each glass, the spectra were obtained from fragments from the interior of the glasses. Thus, one can assume that the glasses possess great homogeneity and, based on the detection of similar signals in the four glasses, it can be assumed that the same type of vitrification was carried out for the G₁–G₄ glasses.

4. Conclusions

By use of a ternary system of the V₂O₅, CdO, and ZnO oxides, novel glassy matrices have been fabricated from unexpected nominal mixture compositions. The IR spectroscopy showed that CdV₂O₆, Cd₂V₂O₇, ZnV₂O₆, and Zn₂V₂O₇ compounds, along with unmelted

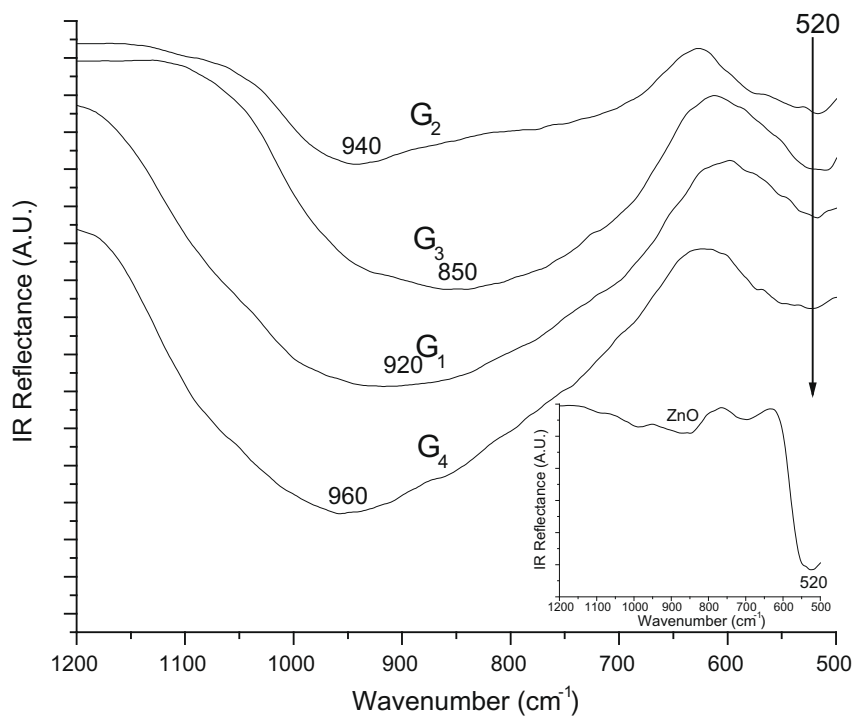


Fig. 2. IR spectra of the G_1 – G_4 glasses and pure ZnO used as a starting material (inset).

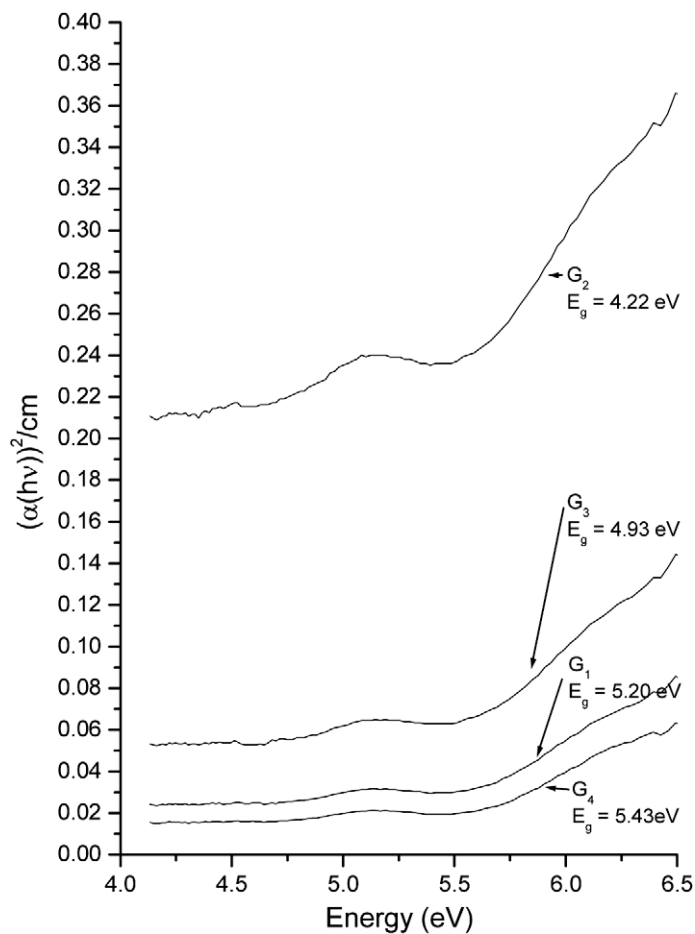


Fig. 3. Optical absorption for G_1 – G_4 glasses.

ZnO residues, were detected in the fabrication of these matrices. A great homogeneity in these glassy matrices was observed since the

micro Raman spectra showed the same signal at different points throughout the interior of each of the glasses.

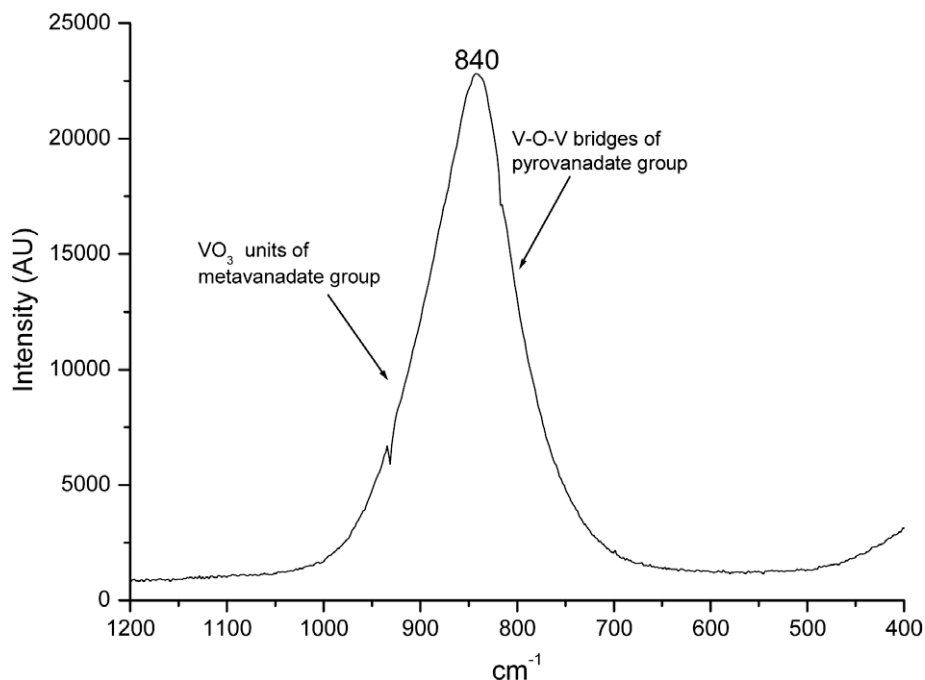


Fig. 4. Representative Raman shift spectrum for G_1 – G_4 glasses measured with the 514 nm line of an argon ion laser.

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